

Project 3: Power System Simulator GUI

Introduction

Project 3 builds on Project 2's solver by adding an interactive GUI based off the elements incorporated in the simulator and Powerworlds simulation model. Rather than building a circuit through a text menu, the user can now drag and drop components directly onto a schematic canvas, wire them together visually, and launch the same Newton-Raphson power flow and symmetrical fault solvers from Project 2 with a single button click. The interface is built using PyQt6 and is organized around the same class structure as before.

How It Works

The simulator keeps the same methodology from Project 2 keeping the Circuit, Bus, Generator, Load, TransmissionLine, Transformer, Jacobian, PowerFlow, and Solution classes unchanged. Each component type now has a visual class including BusVisual, GeneratorVisual, LoadVisual, TransmissionLineVisual, and TransformerVisual. These visuals live on a QGraphicsScene canvas and stay linked to their data objects, so moving a bus on screen automatically redraws every line and transformer connected to it. The CircuitController class bridges the visual layer and the solver, calling the same Solution class methods from Project 2 and routing results into the results panel.

How to Use the Simulator

The simulator launches by running MainWindow.py. The window is divided into three panels: a component palette on the left, a schematic canvas in the center, and a results panel on the right.

Step 1: Place Buses

Drag the Bus button from the left palette and drop it anywhere on the canvas grid. A properties dialog will open automatically, asking for a bus name, nominal kV, and bus type (Slack, PV, or PQ). Fill in the fields and click OK. The bus will appear on the canvas as a labeled horizontal bar.

Step 2: Add Network Elements

Drag a Transmission Line or Transformer button onto the canvas. The properties dialog will ask for a name, a from-bus, a to-bus, and the per-unit R and X values. Transmission lines also ask for shunt G and B. Once confirmed, a line or double-circle transformer symbol will be drawn between the two selected buses and will follow them if either bus is moved

Step 3: Add Generators and Loads

Drag a Generator or Load onto the canvas. The dialog will ask which bus to attach it to, along with the same electrical parameters as Project 2. The symbol will appear directly below its bus and move with it.

Step 4: Run Analysis

Use the toolbar at the top of the window to run the solvers. Click the green button next to "Power Flow" to run the Newton-Raphson solver. Results appear in the right-hand panel showing voltage magnitude, angle, and real and reactive power injection for every bus. Click the orange button next to "Fault Study" to open a small dialog where you select the faulted bus and enter the pre-fault voltage. The panel will then display the fault impedance, subtransient fault current, and post-fault bus voltages.

Step 5: Start a New Circuit

Click the blue button on the far right of the toolbar to clear the canvas and start a fresh circuit. The results panel clears as well.